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Column Assigned: V82

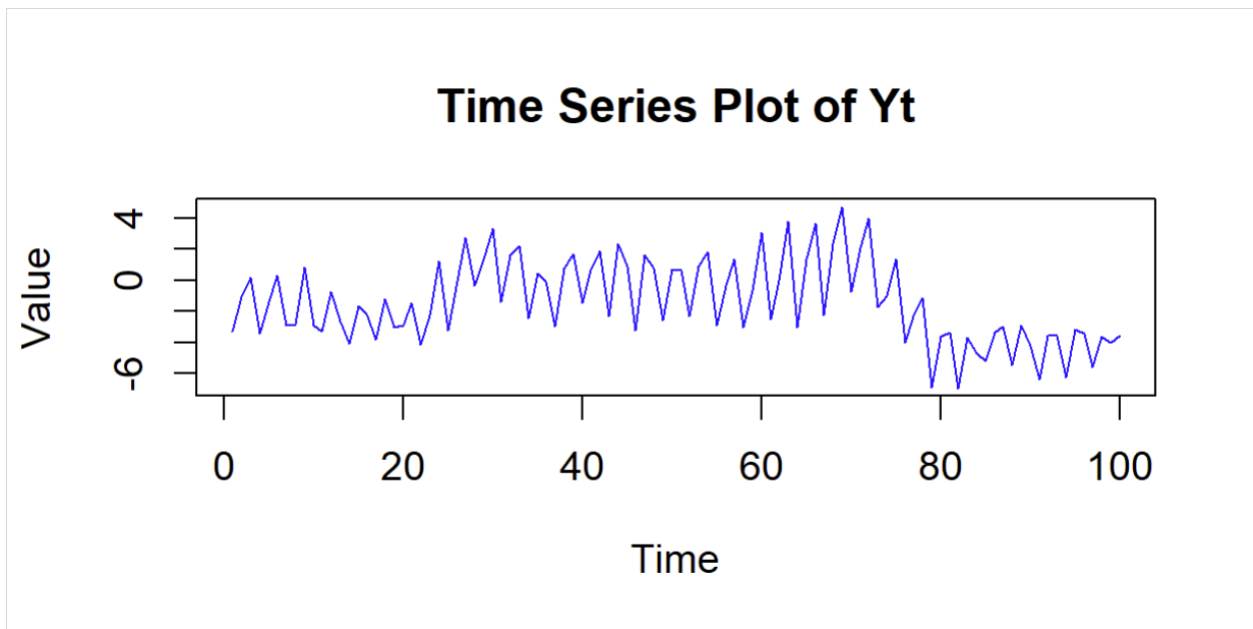
Time Series Analysis

DATA.STAT.780

Assignment 1

Step 1: Preliminary analysis of Orders

1. Plot the times series (Y_t).



#There's a general upward trend in the data points up to around datapoint 70 but after this the trend goes down. There is possible seasonality though it's not quite clear

2a1). The ACF of the original series Y_t shows a high positive autocorrelation at the first lag. The autocorrelations remain statistically significant for many lags (up to lag 19).

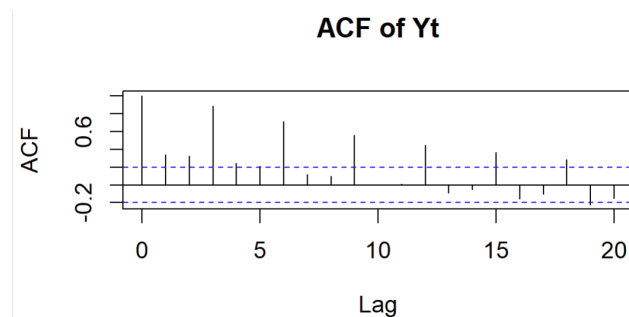


Figure1.1: Original Y_t series

First differencing, There is a strong spike at lag 1 and the correlations still remain significant for many lags, this implies that the **first differencing did not fully remove non-stationarity**.

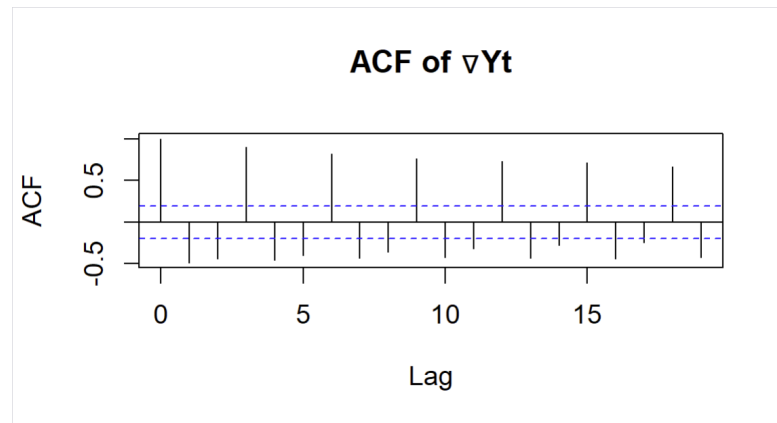


Figure1.2: First differencing

Second differencing, significant spikes remain, still strong negative autocorrelation at lag 1.

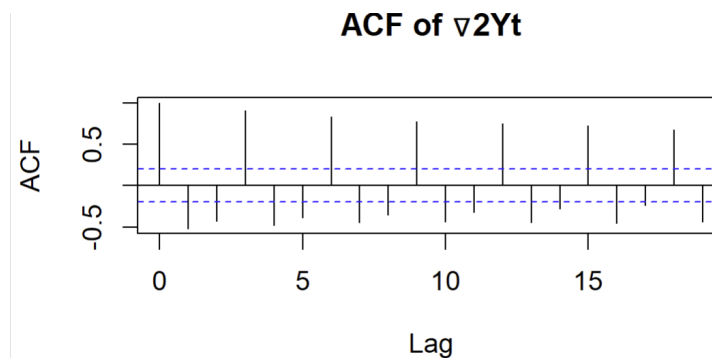


Figure1.3: Second differencing

2a2). The autocorrelations remain large and significant across many lags, which provides graphical evidence of **non-stationarity**.

2b). Augmented DickeyFuller test (ADF)

- The test statistic(ADF test) of the process is **-2.3389** with a **p-value** of **0.4363**. - Greater than **0.05**
- The first-differenced series yields a test statistic of **-3.4099** with a **p-value** of **0.057** - greater than **0.05**
- The first-differenced series yields a test statistic of **-6.8182** with a **p-value** of **0.01** - less than **0.05**

2c)

- Although the ADF test on the first-differenced series yields $p=0.057$, which marginally fails at the conventional 5% level, it passes at the 10% level. Given the small sample size ($n=100$) which reduces ADF power, the near-borderline result, and the risk of over-differencing introduced by $d=2$, we proceed with $d=1$

$d=1$

3) Analysis of results for the first differencing:

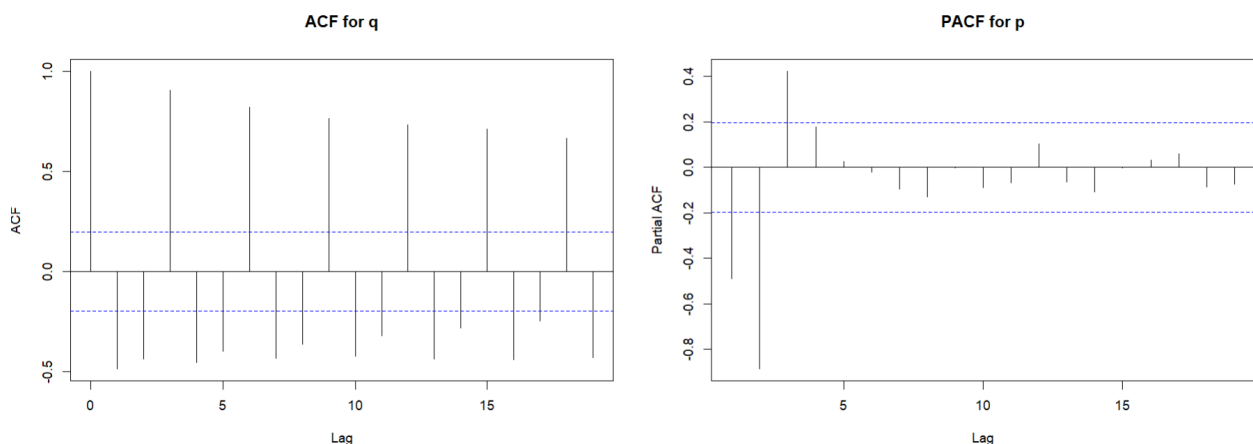


Figure 1.4: plot of ACF and PACF of first differencing

$P_{max} - 3$ - due to a significant spike at lag 3

$q_{max} - 1$ - just slowly decays off so i decide to take 1

Step 2: Estimation and selection of ARIMA Models

1.

	p	q	AIC	BIC
1	1	1	433.6167	441.4020
2	1	2	426.4961	436.8766
3	1	3	379.7337	392.7093
4	1	4	356.9420	372.5127
5	2	1	307.7683	318.1488
6	2	2	301.1361	314.1117
7	2	3	300.4551	316.0259
8	2	4	300.9163	319.0821
9	3	1	297.7128	310.6884
10	3	2	298.0949	313.6656
11	3	3	299.8743	318.0402
12	3	4	301.0282	321.7891
13	4	1	298.4558	314.0266
14	4	2	298.1727	316.3385
15	4	3	299.2060	319.9669
16	4	4	298.6205	321.9766

Figure 2.1: Result table for AIC and BIC for the 16 remaining models.

AIC(The full AIC/BIC table for all 16 models is provided in Figure 2.1)

	p	q	AIC	BIC
9	3	1	297.7128	310.6884
10	3	2	298.0949	313.6656
14	4	2	298.1727	316.3385

BIC

	p	q	AIC	BIC
9	3	1	297.7128	310.6884
10	3	2	298.0949	313.6656
13	4	1	298.4558	314.0266

2.

Among the 16 estimated models, **ARIMA(3,1,1)** minimizes both AIC (297.7128) and BIC (310.6884).

Two adjacent models, ARIMA(3,1,2) and **ARIMA(4,1,1)**

Model	AR Coefficients	MA Coefficients	AIC	BIC	RMSE
ARIMA(3,1,1)	-0.2554, -0.2219, 0.6852	-0.3443	297.71	310.69	1.0087
ARIMA(3,1,2)	-0.3225, -0.3086, 0.6109	-0.3267, 0.1552	298.09	313.67	1.0001
ARIMA(4,1,1)	-0.6194, -0.3589, 0.5679, 0.2232	-0.0196	298.46	314.03	1.0021

Step 3: Diagnostic tests with in-sample data

We test using Ljung–Box test

ARIMA(3,1,1)-output

Result: p-value > 0.05 → fail to reject H_0 -good model

```
Box-Ljung test
data: residuals(model1)
x-squared = 5.7853, df = 10, p-value = 0.833
```

figure 3.1: Ljung–Box test for ARIMA(3,1,1)

ARIMA(3,1,2)-output

Result: p-value > 0.05 → fail to reject H_0 -good model

Box-Ljung test

```
data: residuals(model2)
X-squared = 4.299, df = 10, p-value = 0.9329
```

Figure 3.2: Ljung-Box test for ARIMA(3,1,2)

ARIMA(4,1,1)-output

Result: $p\text{-value} > 0.05 \rightarrow$ fail to reject H_0 - good model

Box-Ljung test

```
data: residuals(model3)
X-squared = 4.8314, df = 10, p-value = 0.9021
```

Figure 3.3: Ljung-Box test for ARIMA(4,1,1)

Conclusion: A p-value greater than 0.05 indicates a failure to reject the null hypothesis (H_0), which suggests that the residuals are consistent with white noise (a desirable outcome).

Now we plot ACF and PACF of residuals

ARIMA(3,1,1)-**output**- spike are inside confidence intervals

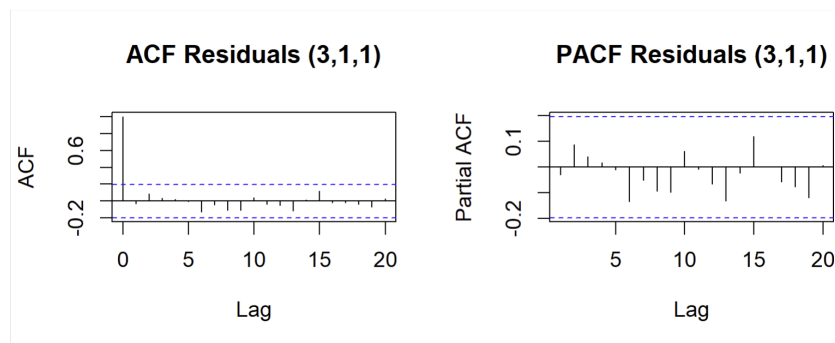


Figure 3.4: plot ACF residuals of ARIMA(3,1,1)

ARIMA(3,1,2)-**output**- spike are inside confidence intervals

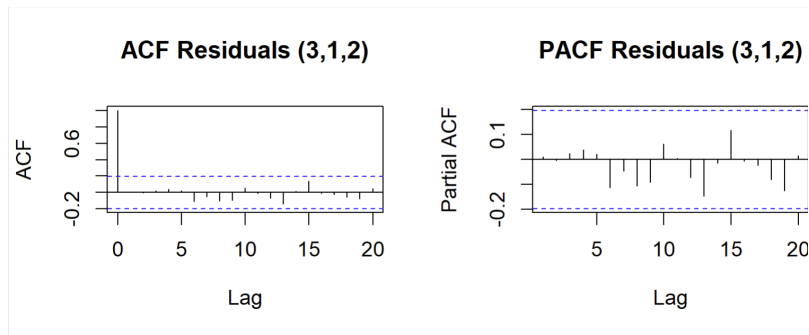


Figure 3.5: plot ACF residuals of ARIMA(3,1,2)

ARIMA(4,1,1)-**output**- spike are inside confidence intervals

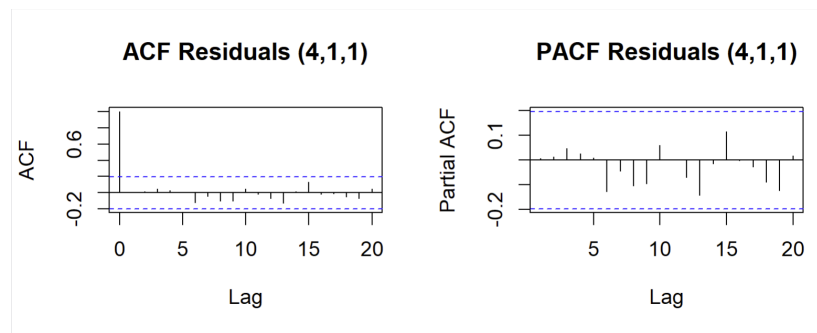


Figure 3.6: plot ACF residuals of ARIMA(4,1,1)

Conclusion: All spikes for our models are inside blue confidence bands implying all our models are good

2.

Here Histogram of plots are presented

Histogram plots for residuals of models ARIMA(3,1,1), ARIMA(3,1,2), ARIMA(4,1,1) respectively

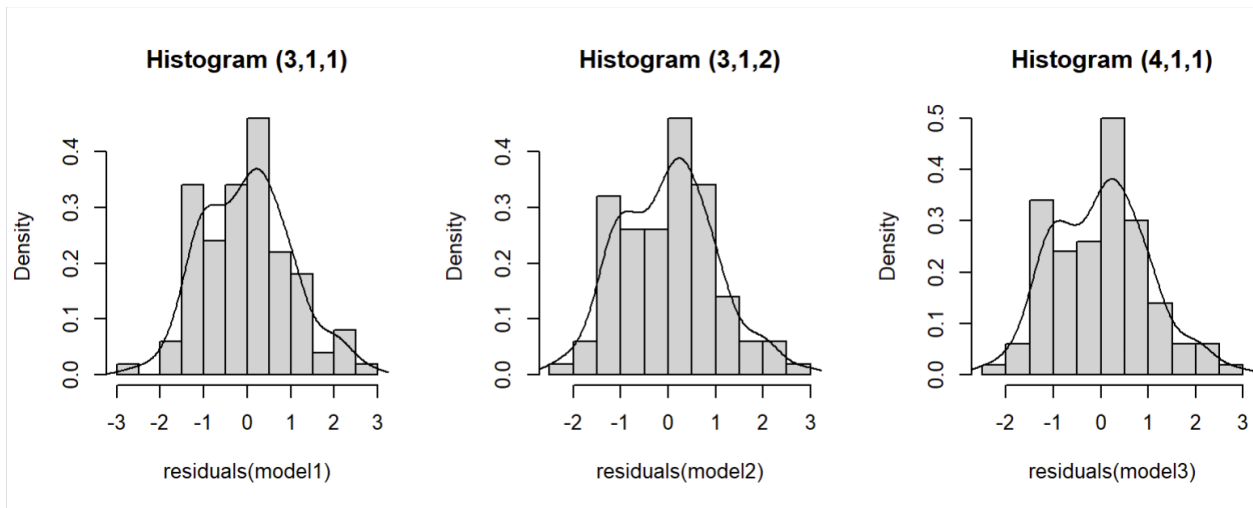


Figure 3.7: Histogram plots for residuals

Conclusion: They are approximately bell shaped implying normality

QQ plots

QQ plots for residuals of models ARIMA(3,1,1), ARIMA(3,1,2), ARIMA(4,1,1) respectively

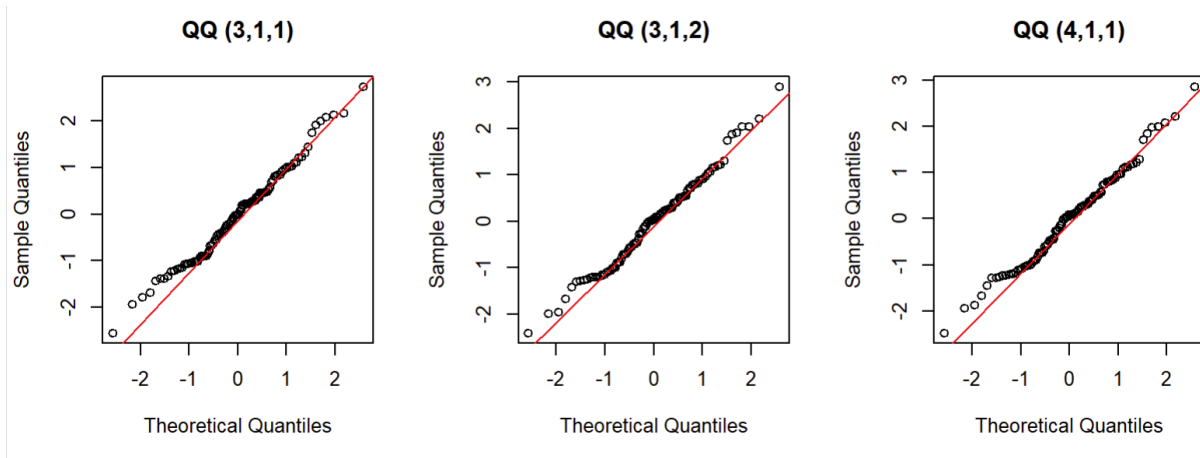


Figure 3.8: QQ plots for residuals

Conclusion: The Points are close to the straight line which means all the models are normal.

Shapiro–Wilk test for ARIMA(3,1,1),

```
shapiro-wilk normality test

data:  residuals(model1)
W = 0.98792, p-value = 0.5027
```

Figure 3.8: Shapiro–Wilk test for ARIMA(3,1,1)

Shapiro–Wilk test for ARIMA(3,1,2)

```
shapiro-wilk normality test

data:  residuals(model2)
W = 0.98696, p-value = 0.4347
```

Figure 3.7: Shapiro–Wilk test for ARIMA(3,1,2)

Shapiro–Wilk test for ARIMA(4,1,1)

```
shapiro-wilk normality test

data:  residuals(model3)
W = 0.98699, p-value = 0.4366
```

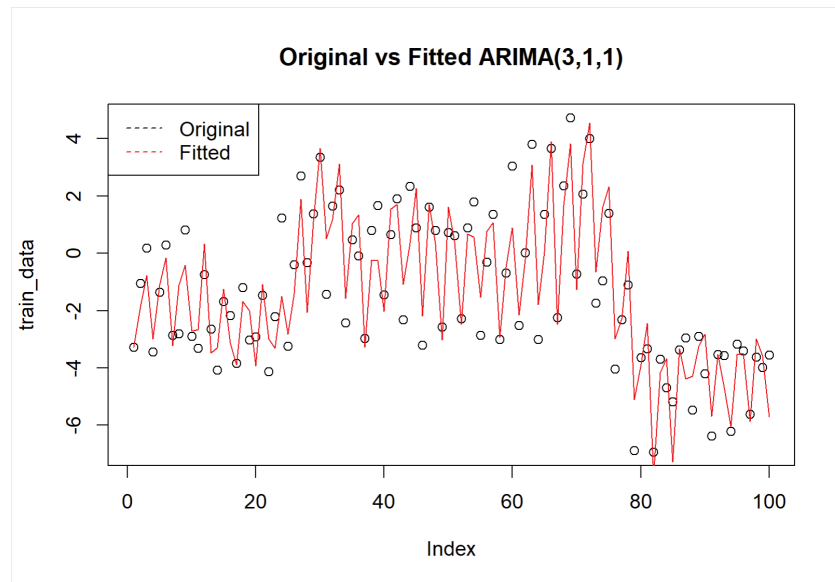
Figure 3.7: Shapiro–Wilk test for ARIMA(4,1,1)

For the **Shapiro–Wilk test** if $p\text{-value} > 0.05 \rightarrow$ residuals are normal. And all the best 3 models have normal residuals.

3.

The best model is ARIMA(3,1,1) because it has the least parameters and fewer MA terms compared to the other 2 competing models.

4.

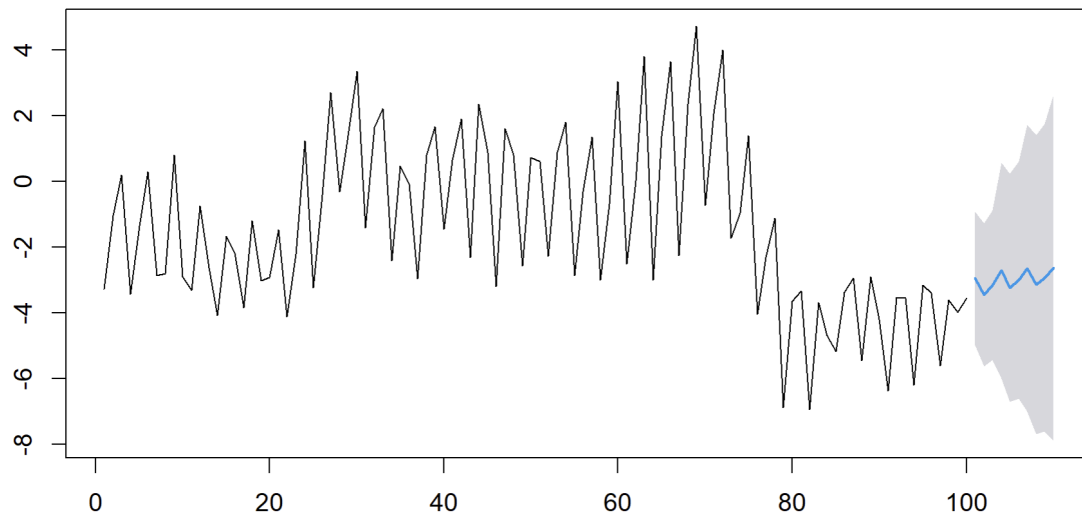


#Fitted the ARIMA(3,1,1) to the data. The fitted values closely track the observed series, indicating that the ARIMA(3,1,1) model captures the main dynamics of the data. Although extreme fluctuations are slightly smoothed, there's no systematic bias evident, and the model appears to provide an adequate in-sample fit.

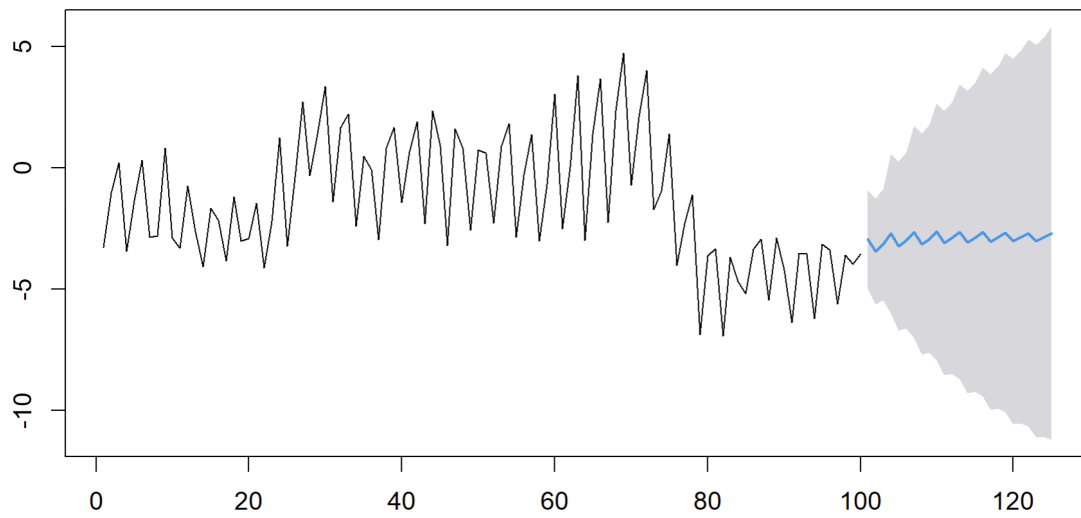
Step 4: Forecast with out-of-sample data

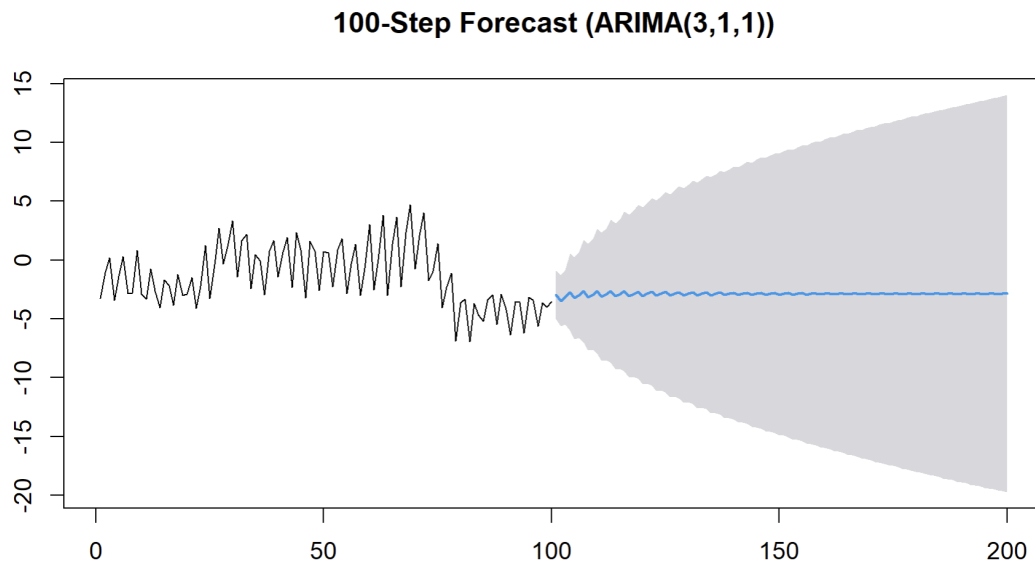
Below are the results plots for the forecast plot for a time series, incorporating a 95% confidence interval.

10-Step Forecast (ARIMA(3,1,1))



25-Step Forecast (ARIMA(3,1,1))





The **10-step** ahead forecast closely follows the recent dynamics of the series. The confidence intervals are relatively narrow, indicating low forecast uncertainty in the short run compared to the 25-step and 100-step.

2.

Mean square Errors

$$1.97 < 3.84 < 10.64$$

MSE increases as forecast horizon increases implying that the model performs well in the short run, but long-run forecasts are subject to high uncertainty